

Assignment1

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Question 4

set up

```
set.seed(7011)
data("iris")
X <- iris[iris[,5]=="setosa", 1]
Y <- iris[iris[,5]=="versicolor", 1]
```

a) Test the normality of X.

```
require(tseries)
```

Loading required package: tseries

```
Registered S3 method overwritten by 'quantmod':
  method      from
as.zoo.data.frame zoo
```

```
jarque.bera.test(X)
```

Jarque Bera Test

```
data: X
X-squared = 0.36208, df = 2, p-value = 0.8344
```

The hypothesis test is $H_0 : X$ is normal against $H_a : X$ is not normal . We run Jarque-Bera test to check the normality. Since the reported p-value is $0.8344 > 0.05$, we don't have enough evidence to reject H_0 on the 0.05 confidence level. We don't reject X is normal.

b) Assuming $\sigma_x = \sigma_y$, test $\mu_x = \mu_y$ against $\mu_x \neq \mu_y$.

```
t.test(X,Y,var.equal = TRUE)
```

Two Sample t-test

```
data: X and Y
t = -10.521, df = 98, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -1.1054165 -0.7545835
sample estimates:
mean of x mean of y
  5.006      5.936
```

The hypothesis test is $H_0 : \mu_x = \mu_y$ against $H_a : \mu_x \neq \mu_y$. $\sigma_x = \sigma_y \implies \sigma_x^2 = \sigma_y^2 = \sigma^2$. We run Two sample t-test to check if the true means of X and Y are same. Since the reported p-value is extremely small and less than 0.05, we have enough evidence to reject H_0 on the 0.05 confidence level. True difference in means is not equal to 0.

c) Assuming $\sigma_x \neq \sigma_y$, test $\mu_x \leq \mu_y$ against $\mu_x > \mu_y$.

```
t.test(X,Y,var.equal = FALSE,alternative = "greater")
```

Welch Two Sample t-test

```
data: X and Y
t = -10.521, df = 86.538, p-value = 1
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 -1.07697      Inf
sample estimates:
mean of x mean of y
  5.006      5.936
```

The hypothesis test $H_0 : \mu_x \leq \mu_y$ against $H_a : \mu_x > \mu_y$. $\sigma_x \neq \sigma_y \implies \sigma_x^2 \neq \sigma_y^2$. We run Welch two sample t-test to check if the true means of X is less than or equal to that of Y . Since the reported p-value is $1 > 0.05$, we don't have enough evidence to reject H_0 on the 0.05 confidence level. We don't reject $\mu_x \leq \mu_y$.

d) Test $\sigma_x^2 \geq \sigma_y^2/2$ against $\sigma_x^2 < \sigma_y^2/2$.

```
var.test(X,Y,ratio = 2, alternative = "less")
```

F test to compare two variances

data: X and Y

F = 0.23317, num df = 49, denom df = 49, p-value = 5.568e-07

alternative hypothesis: true ratio of variances is less than 2

95 percent confidence interval:

0.0000000 0.7495481

sample estimates:

ratio of variances

0.4663429

The hypothesis test is $H_0 : \sigma_x^2 \geq \sigma_y^2/2$ against $H_a : \sigma_x^2 < \sigma_y^2/2$. We run F test to compare two variances. Since the reported p-value is $5.568e-7 < 0.05$. We have enough evidence to reject H_0 on the 0.05 confidence level. The variance of X is less than the half variance of Y .